PNRR-PNC · TOR VERGATA INNOVATION DAY 2026

SUPER: from the programmable data plane to the sovereign cloud

Open-source components for high-availability cloud/edge infrastructures with GPU sharing

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PNRR · RESTART · Spoke 4 · Sub-project S2 SUPER

The challenges

1 **Expensive, underused GPUs.**
Static assignment to VMs and containers.

2 **Lock-in.**
Licensing costs of proprietary platforms.

3 **Reliable measurements.**
Performance in virtualized environments.

4 **Per-flow monitoring.**
Fine-grained traffic metrics, without overloading the nodes.

FIGO Federated Infrastructure for GPU Orchestration

Today

- **OpenStack-like managers assign GPUs statically**
- **An allocated GPU stays locked to its VM**
- **...even when the VM is not running**

With FIGO

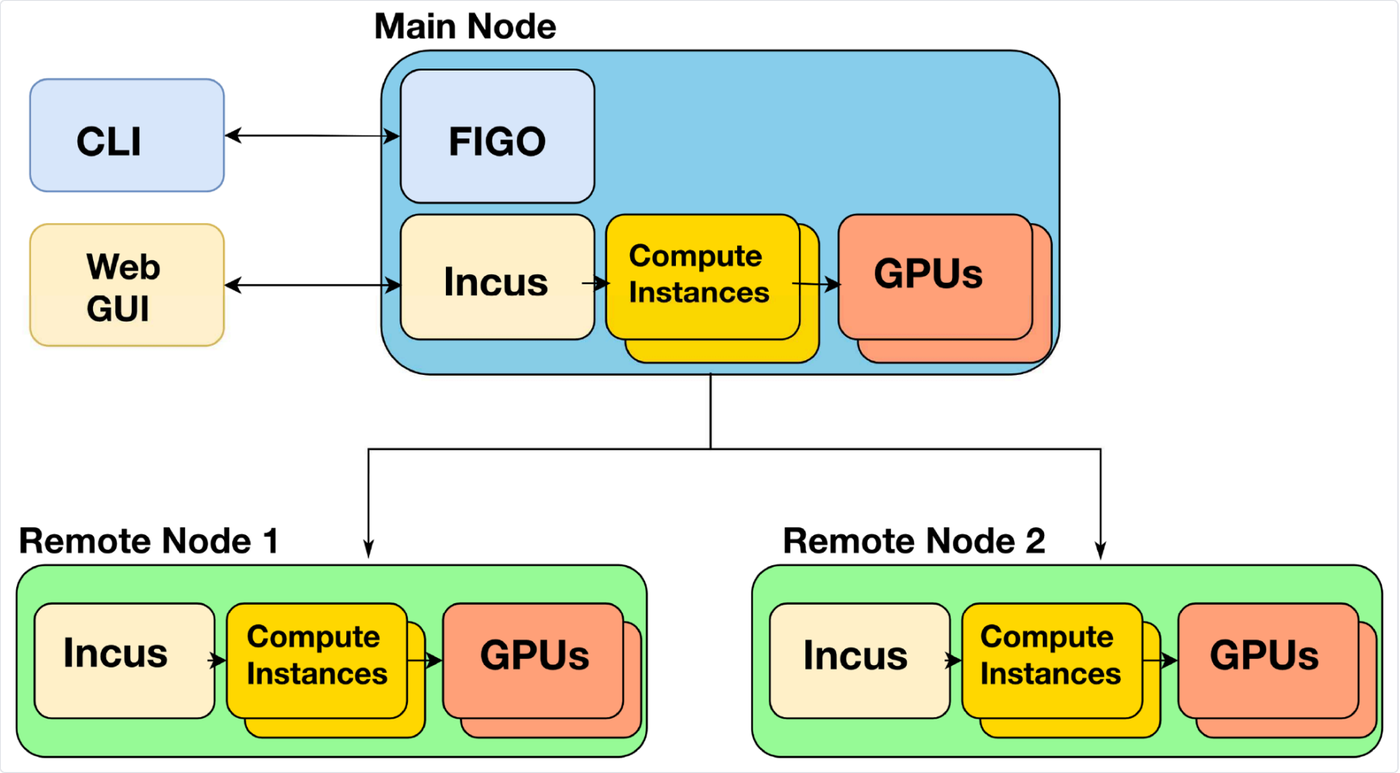
- **Dynamic** GPU allocation to VMs and containers
- Owners keep **priority** on their GPUs
- Idle GPUs are **shared** across the federation

Takeaway. Higher GPU utilization, and GPU access for researchers who have none.

FIGO — architecture

- Built on **incus**
- Manages VMs and containers
- GPU profiles and users
- Federated remote nodes

Open source:
github.com/StefanoSalsano/figo



Each remote node runs incus, coordinated by the main node

FIGO — command line LIVE DEMO

```
# Create a new user
figo user add john_doe -e john@example.com -n "John Doe" \
     -o "Example Corp" -w -k -s

# Enroll a remote node
figo remote enroll my_remote 192.168.1.100 8443 ubuntu \
     ~/.config/incus/client.crt /var/lib/incus/server.crt --loc_name main

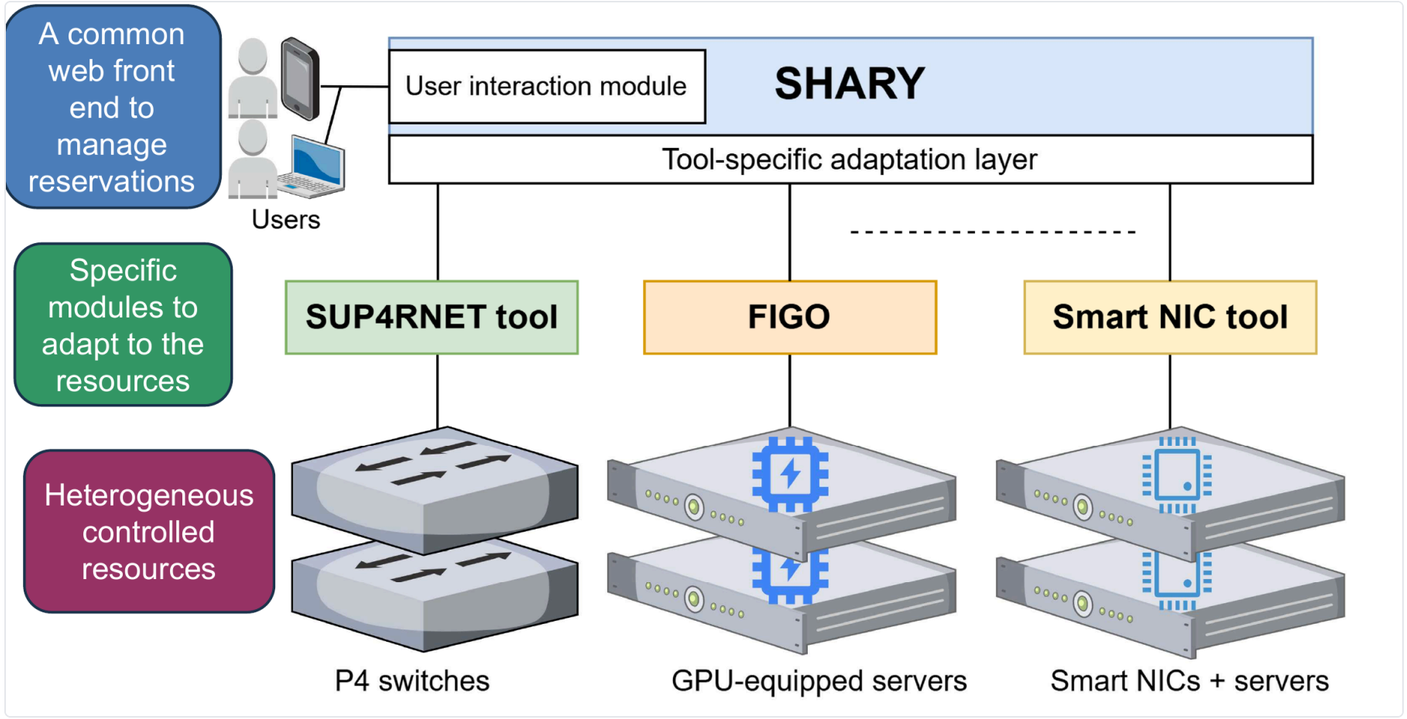
# Attach a GPU to an instance on a remote node
figo gpu add my_remote:my_project.instance_name
```

One CLI for instances, GPU profiles, users and remote nodes.

SHARY SHaring Any Resource made easY

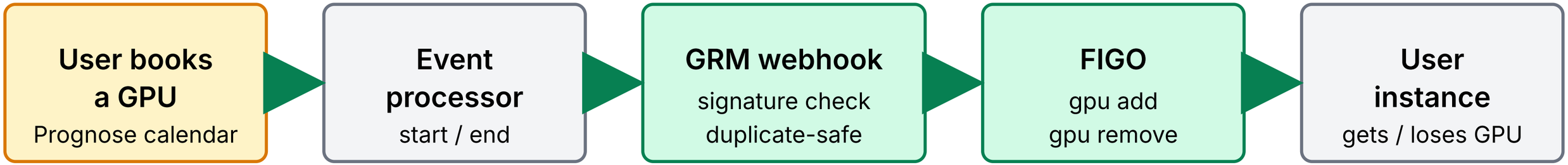
- **One web front end**: Prognose
- to book RESTART testbed resources:
- GPUs, SmartNICs, P4 switches
- Calendar booking, conflict checks
- Specific module per resource

IEEE NOMS 2025 — *Sharing GPUs and Programmable Switches in a Federated Testbed with SHARY*



Common front end and specific modules for each resource

From booking to GPU: Prognose + FIGO



- **GRM** — GPU Reservation Manager, an external webhook
- Reservation intent (Prognose) separated from enforcement (FIGO)
- Resilient to duplicate events, GRM restarts, FIGO unavailability

Takeaway. Enforceable GPU reservations, without modifying the reservation platform.

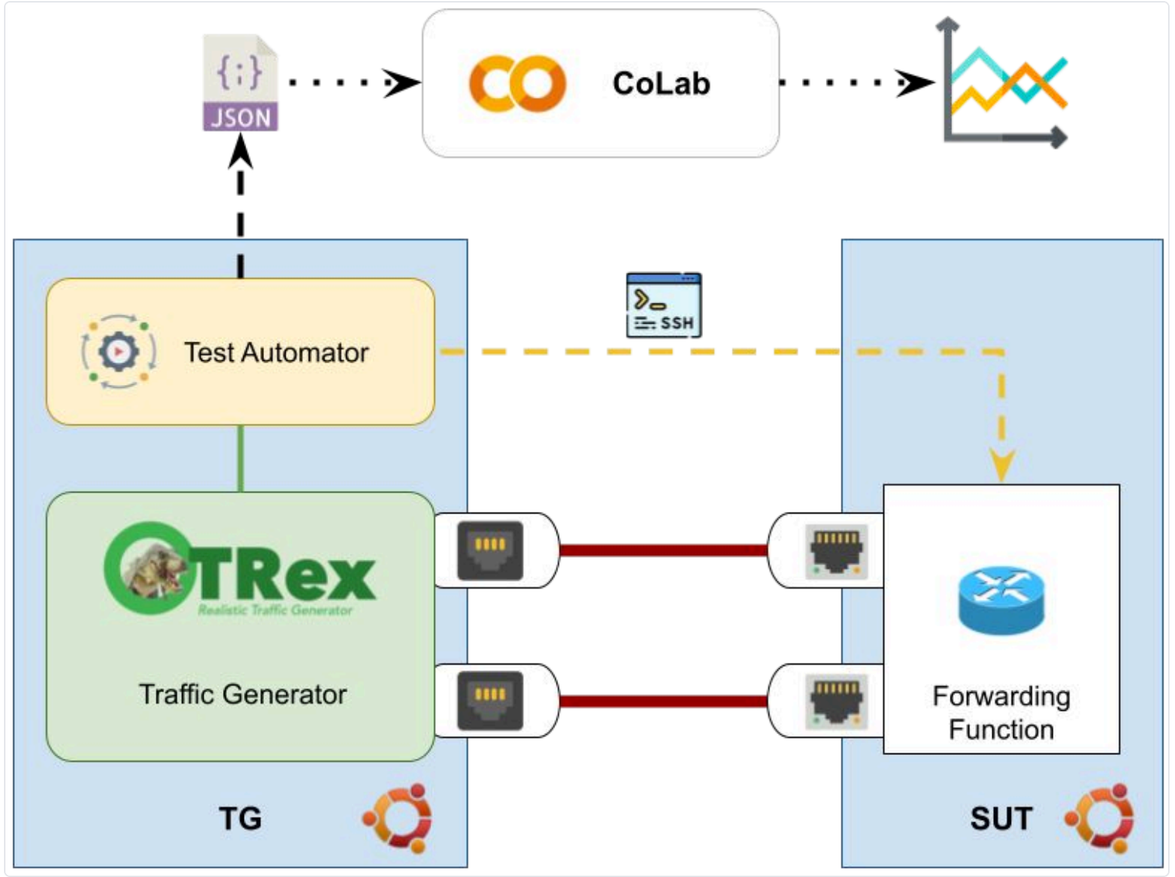
E. Yazdani, MSc thesis, University of Rome Tor Vergata, A.Y. 2024–25 · Prognose: docs.prognose.crownlabs.polito.it

PASTRAMI — trustworthy performance measurements

Performance **A**ssessment of **S**of**T**ware **R**outers **A**ddressing **M**easurement **I**naccuracy

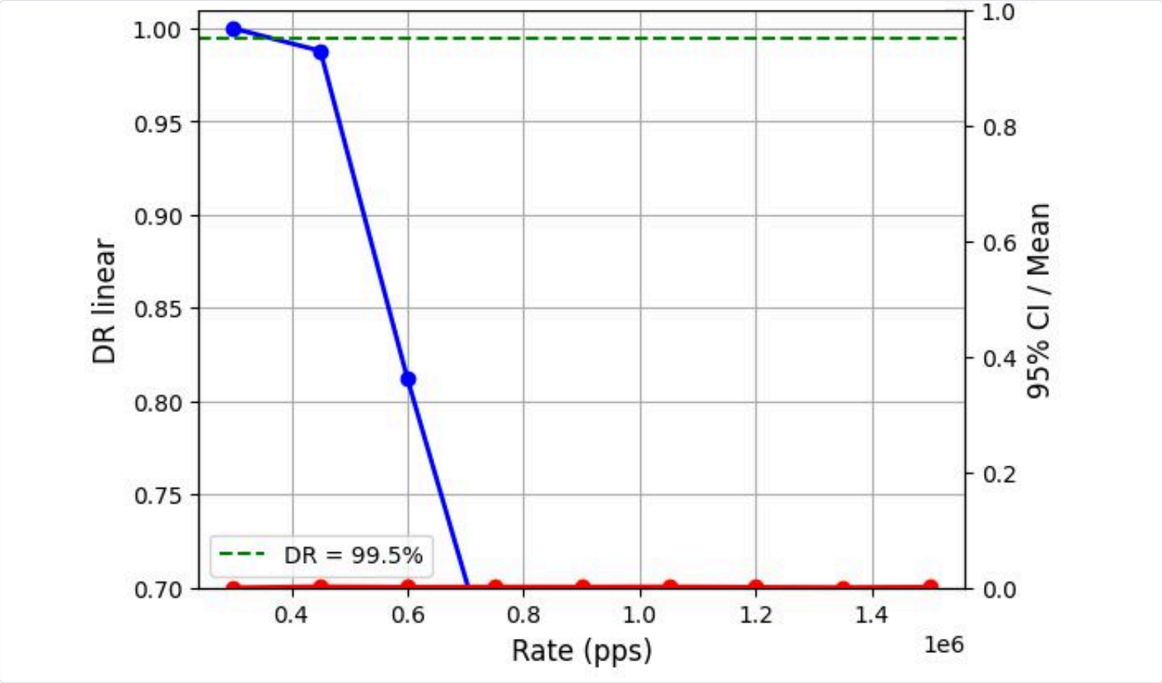
- Traffic generator → **system under test**
- Linux software routers (e.g., SRv6)
- Scans a **wide range of rates**
- Loss and delivery curves, 95% CI
- Is **PDR@0.5%** reliable here?

PDR@0.5% = Partial Drop Rate: max rate with ≤0.5% loss

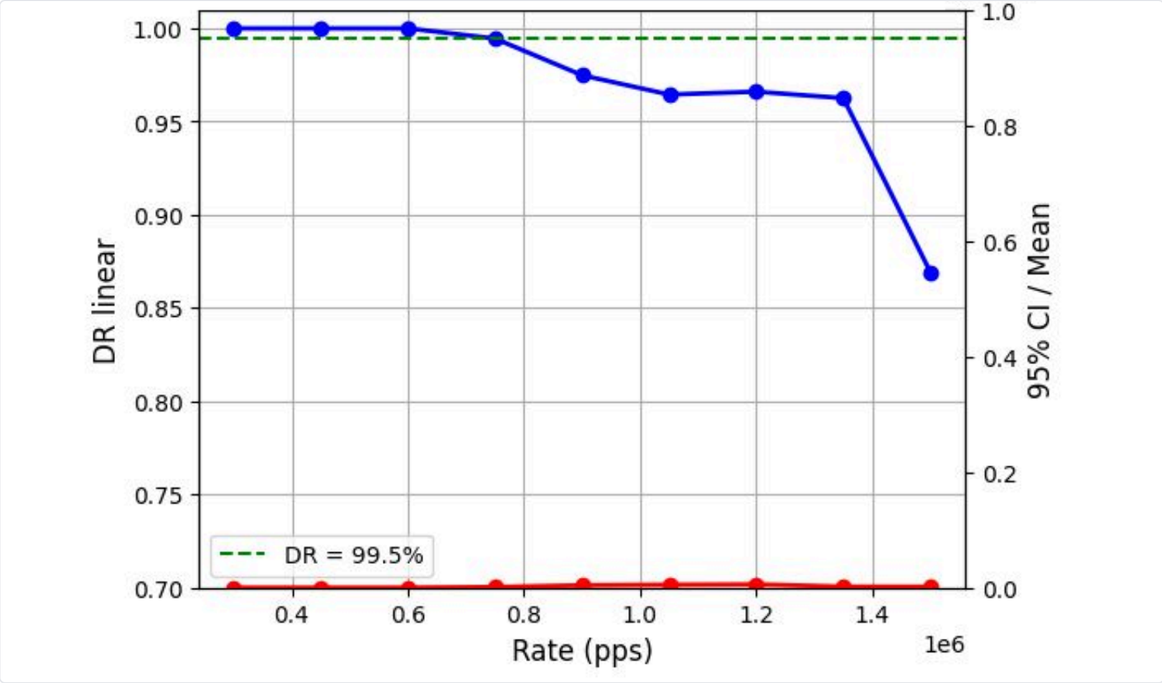


Test automator, TRex traffic generator, system under test, Colab post-processing

PASTRAMI — is the measurement reliable?



Good: sharp transition to saturation → PDR@0.5% is reliable

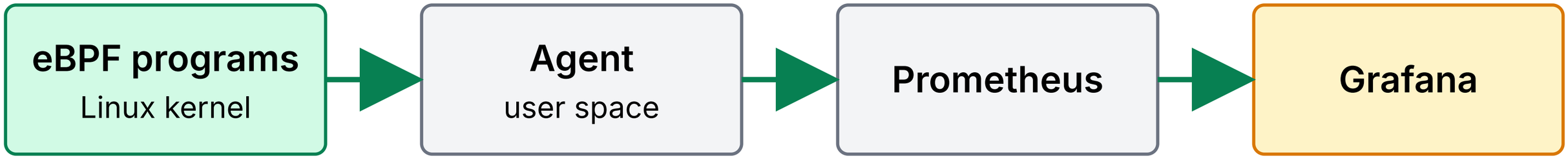


Bad: gradual losses → PDR@0.5% alone is misleading

- Saturation detected via the **Throughput Increase Ratio**
- **netrace**: eBPF CPU-load tracing — Linux kernel versions behave differently

Open source: github.com/netgroup/pastrami

eFFECT — per-flow monitoring with eBPF



- Flow-level metrics collected **in the kernel**
- **High temporal granularity**, minimal overhead

Takeaway. Per-flow metrics, from the kernel to Grafana dashboards.

Towards the sovereign cloud

An **open-source**, **high-availability** cloud/datacenter platform, sovereign and lock-in free, with **GPU sharing** for AI workloads.

CloudStack · Linux · eBPF · SRv6 · GPU sharing

Platform base TRL 9

CloudStack in high availability, in operation in Common Net datacenters.

SUPER components

Open-source prototypes FIGO, SHARY, PASTRAMI, eFFECT converging into the platform.

Also in SUPER: XDP-accelerated IDS (Suricata); adaptive SRv6 paths for LLM training.

Nano-f — from research to market

Local and sovereign infrastructures, designed with hyperscale efficiency principles

- **Spin-off** of the University of Rome Tor Vergata, approved, being incorporated
- Industrial and commercial partner: **Common Net**, cloud provider and ISP
- Private and sovereign clouds, edge cloud, **GPU-as-a-Service**
- Advanced networking: Linux, eBPF, SRv6
- For ISPs, cloud providers, enterprises and public administration

Consulting → Managed services → Software platform → Licenses

Let's meet

B2B meetings

12:00–15:15 · Room L4

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FIGO on GitHub



SUPER on RESTART

Thank you.